

Programme	B. Sc. Mathematics Honours			
Course Code	MAT4FV110(2)			
Course Title	THE MATHEMATICAL PRACTICES OF MEDIEVAL KERALA			
Type of Course	VAC			
Semester	IV			
Academic Level	200 - 299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	3	3	-	45
Pre-requisites	1. Fundamental Mathematics Concepts: Number system, Basic Mathematical operations, Plane Geometry. 2. Convergence of series of numbers and functions.			
Course Summary	This course familiarises students with the traditional Indian Mathematics practised in the Medieval Kerala School of Astronomy and Mathematics.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Uncover the underlying fundamental principles of the traditional mathematics practised in medieval Kerala.	U	C	Seminar Presentation/ Group Tutorials
CO2	Appreciate the role of thought process and working rules in mathematics.	U	C	Seminar Presentation/ Group Tutorials
CO3	Appreciate the usage of infinite series in mathematical analysis.	U	C	Seminar Presentation/ Group Tutorials
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		1. Lilavati of Bhaskaracarya Translated by K.S.Patwardhan, S.A.Naimpally and S.L.Singh, Motilal Banarsidass Publishers, Delhi. 2006. 2. Ganita Yukti Bhasa of Jyesthadeva. Volume I. English Translation by K.V.Sarma with explanatory notes by K.Ramasubramanian, M.D.Srinivas and M.S.Sriram. Hindustan Book Company, 2008.		
Module	Unit	Content	Hours (36 + 9)	Ext. Marks (50)
I	Measurement of sides and areas of triangles, quadrilaterals and circles.		9	14
	1	Computation of sides of a right triangle when one side is given.		
	2	Computation of area of triangles and quadrilaterals.		
	3	Computation of the perpendicular below the intersection of diagonals.		
	4	Approximating the surface area and volume of spheres.		
	5	Computation of sides of polygons inscribed in a circle.		
	6	Computation of the arcs and chords of circles.		
	Chapter 28 from Text I (Treatment based on English translations of Sanskrit verses in Lilavati).			
II	Rules concerned with Solids, Shadow of Gnomon and Pulverizer.		9	12
	7	Volume of Solids		
	8	Volume of a heap of Grain		
	9	Shadows of Gnomon.		
	10	Pulverization		
	Chapters 29, 30, 31, 32 and 33 from Text I (Treatment based on English translations of Sanskrit verses in Lilavati).			
III	Circle and Circumference as in Yuktibhasa.		10	14
	11	Circumference of a circle approximated by regular polygons.		
	12	Circumference of a circle without calculating square roots.		
	13	Circumference of a circle in terms of the hypotenuses.		
	14	Summation of Series.		
	15	Calculation of circumference.		
	16	Conversion of the Rsine to Arc.		
	Sections 6.1 to 6.6 of Chapter 6 from Text II.			
IV	Sine and Cosine series as in Yuktibhasa.		8	10
	17	Some technical terms and derivation of Rsines.		
	18	Computation of Rsines.		
	19	Computation of Jya and Sara by sankalita and accurate circumference.		
	Sections 7.1 to 7.6 of Chapter 7 from Text II.			
V (Open Ended)	From Ancient Mathematical Rules to Modern Computer Algorithms.		9	
	20	Decoding of important Sanskrit verses discussed in Modules I and II from Lilavati (Text I).		

21	Decoding of important Sanskrit verses discussed in Modules III and IV from Yuktibhasha (Text II).		
22	Conversion of selected Rules discussed in Modules I to IV into Computer Algorithms.		
Relevant Topics from Text I, Text II and References.			

References:

1. The Mathematics of India - Concepts, Methods, Connections. P.P.Divakaran, Hindustan Book Agency, New Delhi, 2018.
2. A Passage to Infinity - Medieval Indian Mathematics from Kerala and its Impact. George Ghevarghese Joseph, Sage Publications, New Delhi, 2009.
3. On an Untapped Source of Medieval Keralese Mathematics. C.T.Rajagopal and M.S.Rangachari, Archive for the History of Exact Sciences, 35 (2), (1986), 91 - 99.
4. Yuktibhasha. Rama Varma Maru Thampuram and A.R.Akhilesvara Iyer (Editors)}, Mangalodayam Press, Trichur 1948.
5. Tantrasangraha of Nilakantha Somayaji with Yuktidipika and Laghuvivrti of Sankara. K.V.Sarma, Vishveshvaranand Visva Bandhu Institute of Sanskrit and Indological Studies, Punjab University, Hoshiarpur 1977.
6. Colebrook's translation of the Lilavati with Notes by Haran Chandra Banerji. The Book Company, Calcutta, 1927.
7. Mathematical Treasures – Lilavati of Bhaskara. Frank J.Swetz and Victor J.Katz. Loci. 2011.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	1	1	3	2	1	0	2	3	0
CO 2	2	3	1	2	2	3	1	0	2	3	0
CO 3	2	2	2	2	2	1	1	0	2	2	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

VOCATIONAL MINORS